MILITARY SPECIFICATION

PLASTICS MATERIALS, POLYESTER RESIN, GLASS FIBER BASE, FILAMENT WOUND TUBE

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the requirements for glass fiber base, filament wound polyester resin plastic materials for use in fabrication of tubes.

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-Y-1140	Yarn, Cord, Sleeving, Cloth, and Tape-Glass
MIL-P-9400	Plastic Laminate Materials and Sandwich Construction, Glass Fiber Base, Low Pressure Aircraft Structural, Process Specification Requirement
MIL-R-21607	Resins, Polyester, Low Pressure Laminating, Fire Resistant

STANDARDS

FEDERAL

Fed. Test Method Std. No. 406	Plastics; Methods of Testing

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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ASTM STANDARDS

ASTM-D-2290-64T	Test for Apparent Tensile Strength of Parallel Reinforced Plastics by Split Disk Method (Tentative)
ASTM-D-2291-64T	Recommended Practice for Fabrication of Ring Test Specimens (Tentative)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Preproduction sample. This specification requires the submission of preproduction samples for preproduction testing. See 4.3 for details.

3.2 Materials. Materials shall be as specified herein. When materials not specifically designated are used, such materials shall be subject to the approval of the procuring activity.

3.2.1 Glass fiber continuous roving. Continuous roving shall be a collection of strands gathered without mechanical twist. The roving shall consist of standard K filament strand in 34 ends (MIL-Y-1140, Class C, Form 1, designation ECK). A polyester, compatible silane sizing shall be applied to the roving.

3.2.2 Resin. The resin shall be in accordance with MIL-R-21607; the class of resin shall be specified by the procuring agency.

3.2.3 Fillers, pigments, and other additives. The handling characteristics of the resin may be modified with a maximum of five percent (based upon weight of resin) of fillers or other additives, providing the clarity of the resin and the translucency of the resultant laminate remain unaffected where a Class A resin is specified. Pigments shall be used when specified by the procuring agency.

3.2.4 Gel coat. Use of a gel coat having a maximum thickness of 0.015 inch is permissible unless the contract or order prohibits it. If used, the gel coat shall show no tendency to delaminate, peel, or craze.

3.2.5 Glass cloth. The glass cloth shall be in accordance with MIL-Y-1140, Class C, Form 4, Cloth No. 112. The glass cloth shall be laid upon the gel coat or, as an alternate method, it may be preimpregnated with the polyester resin mix, used during the winding operation, and applied directly to the mold surface of the mandrel.

3.3 Process. A detailed description of the manufacturing and fabrication process and methods of control of manufacturing variables in the form of a titled, numbered, and dated process specification shall be prepared by the prime contractor, or obtained from a subcontractor, and approved by the prime contractor. The process specification shall comply with the requirements of MIL-P-9400 and follow its format as closely as possible. The process specification shall be forwarded by the prime contractor to the procuring activity for review and approval prior to commencement of the production of material to which it applies and, thereafter, during the

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course of production for use by authorized Government and industry inspectors in the facilities of the prime contractor, his subcontractors, or his vendors. It shall also be made available, on request, for review by engineers of the procuring activity. The prime contractor represents that compliance with such process specification will produce a material meeting all the contract requirements. Any changes in the process specification shall be approved by the procuring activity prior to any change in production.

3.4 Mechanical and physical properties

3.4.1 Mechanical and physical properties shall be as specified in Table I. No individual specimen tested for mechanical strength shall indicate a value lower than 80 percent of the average of the results of the 5 specimens tested for the property. No individual sample tested for void content shall exceed the maximum percentage specified in Table I.

3.4.2 Laminates shall be fully cured after the initial curing operation is completed. Flexural strength of post-cured materials shall not exceed the average flexural strength of specimens tested under standard conditions by more than 15 percent (see 4.3.3.1). Failure to meet this requirement will be considered an indication of undercure and the basis for rejection of the preproduction unit.

3.4.3 The average of ten Barcol hardness measurements that are made on production units shall be not lower than the Barcol hardness specified in Table I.

3.5 Workmanship

3.5.1 The laminate furnished under this specification shall be uniformly representative of good commercial practice in appearance and quality.

3.5.2 The allowable defects in the plastic material of the parts shall consist of those included by the contractor as a part of the process specification and then specifically as approved by the procuring activity. These allowable defects shall be fully described in the process specification as to type, size, number, extent, spacing, et cetera.

3.5.3 The reparable defects, if any, in the plastic material of the parts shall consist of those which can be repaired without affecting the serviceability of the part and shall be only as specified in the approved process specification. The reparable defects and repair methods shall be fully described in the process specification. Allowable defects shall not affect the serviceability of the part and shall be specified only in the approved process specification.

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3.5.4 Defects greater than those listed in the process specification as allowable or reparable shall be cause for rejection.

3.5.5 Where requirements relative to defects, as stated in the orders, contracts, drawings, or specifications for specific parts, are more severe than those in the approved process specification, the requirements of the orders, contracts, drawings, or specifications for the specific parts shall govern.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection and tests. The inspections and tests that are conducted on reinforced plastic laminates under this specification shall be classified as follows:

(a) Preproduction unit (see 4.3)

(b) Quality conformance inspection (see 4.4)

4.3 Preproduction unit inspection

4.3.1 Visual and dimensional examination. Each individually laminated part shall be examined for visual (as per 3.5.2) and dimensional characteristics.

4.3.2 Preproduction sample. Test specimens shall be taken from locations on the preproduction unit (see 3.1) as approved by the Government inspector. Where possible, the test samples shall be obtained from areas normally cut out of the part. In the unusual cases where the item is of such a nature that satisfactory samples cannot be obtained, or where removal of such samples and subsequent repair will seriously impair its serviceability, prototype units may be fabricated to the same thicknesses using the same materials and process to be employed under the contract.

4.3.3 Preproduction tests. Preproduction tests shall consist of all tests listed in Table I as well as those described in 4.3.3.1 and 4.3.3.2.

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TABLE I

MECHANICAL PROPERTIES - FIBROUS GLASS, FILAMENT WOUND PLASTIC MATERIALS

Property to be Tested	Direction	Test Method		Specimens		Conditioning Procedure	Unit of Value	Value Required
		Specification FED STD 406	Modified by	Size	Number Tested			
Flexural Strength Ult	Circumferentially	1031	--	Modified by 4.5.1.3.3	5	Standard	P. S. I. Min. Avg.	25,000
Flexural Modulus	Circumferentially	1031	--	"	5	Standard	P. S. I. Min. Avg.	1.1×10^6
Flexural Strength Ult	Longitudinally	1031	--	"	5	Standard	P. S. I. Min. Avg.	22,000
Flexural Strength Ult	Circumferentially	1031	--	"	5	Wet	P. S. I. Min. Avg.	22,000
	and Longitudinally	1031	--	"	5	Wet	P. S. I. Min. Avg.	19,000
Flexural Modulus	Circumferentially	1031	--	"	5	Wet	P. S. I. Min. Avg.	$.95 \times 10^6$
Flexural Strength Ult	Circumferentially	1031	at 160°F	"	5	30 Minutes at 160°F	P. S. I. Min. Avg.	20,000
Tensile Strength Ult	Circumferentially	--	4.5.4	Modified by 4.5.1.3.1	5	Standard	P. S. I. Min. Avg.	20,000
Tensile Strength Ult	Longitudinally	1011	--	Modified by 4.5.1.3.2	5	Standard	P. S. I. Min. Avg.	20,000
Void Content	--	--	4.5.3	2 inches dia.	5	--	Percent Max	2
Resin Content	--	7061	--	2 inches dia.	5	--	Percent Max	40
Barcol Hardness	--	--	4.5.2.2	4.3.2	10	Standard	Barcol	50 minimum

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4.3.3.1 Determination of cure. The degree of cure of the preproduction unit shall be determined by comparing the average flexural strength of five specimens tested in one direction after standard conditioning with the average of five specimens tested after postcure (see 4.5.2.1).

4.3.3.2 Barcol hardness. If the preproduction unit meets the requirements of 3.4.2, ten readings of Barcol hardness are to be made as specified in 4.5.2.2 and the average of the readings shall be not lower than the Barcol hardness specified in Table I.

4.4 Quality conformance inspection and tests

4.4.1 Visual and dimensional examination. Each individually laminated part shall be examined for visual (as per 3.5.2) and dimensional characteristics.

4.4.2 Test lot. A test lot shall consist of the production units offered for delivery at one time or during a fixed period of time, which have been manufactured or processed under substantially the same conditions as the preproduction unit with no change in glass reinforcement or resin formulation.

4.4.3 Sampling for quality conformance tests. A random sample of units shall be selected by the inspector from each lot in accordance with Table II for the tests specified in 4.4.4.1 and 4.4.4.2.

Table II - Sampling of Units for Quality Conformance Tests

Number of Units in Lot	Number of Units in Sample for Tests Specified in:		Rejection Number for Tests Specified in:	
	4.4.4.1	4.4.4.2	4.4.4.1	4.4.4.2
2 to 9	2	2	0	0
10 to 15	2	3	0	0
16 to 25	3	5	0	0
26 to 40	3	7	0	0
41 to 65	5	10	0	1
66 to 110	7	15	0	1
111 to 180	10	25	1	2
181 to 300	15	35	1	3

4.4.4 Quality conformance tests. Specimens from each of the production units in the sample selected in accordance with 4.4.3 shall be subjected to the tests specified in 4.4.4.1 and 4.4.4.2. A failure of the average of the test results to conform with any of the applicable requirements shall be considered a failure of that production unit, and if the number of nonconforming units in any sample exceeds the acceptance

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number in Table II for that lot size, the lot represented by the sample shall be rejected. The lot may be resubmitted only after it has been reworked by the manufacturer to repair or remove and replace nonconforming items to the satisfaction of the inspector.

4.4.4.1 Flexural strength (wet). The test for flexural strength (wet) determined in one direction shall be conducted on sample units selected in accordance with Table II to determine conformance with the requirements shown in Table I.

4.4.4.2 Resin content, percent voids, and Barcol hardness. Tests shall be conducted on samples selected in accordance with Table II to determine conformance with requirements shown in Table I and 3.4.3.

4.5 Test methods

4.5.1 Mechanical and physical tests. Mechanical tests shall be made in accordance with the applicable test methods specified in Table I.

4.5.1.1 Test specimens. The number and size of specimens per sample unit shall be as specified in Table I.

4.5.1.2 Conditioning. Specimens shall be conditioned before test, as specified in Table I.

4.5.1.2.1 Standard condition. Standard condition shall be 23° ± 1°C (73.5° ± 2°F) and 50 ± 4 percent relative humidity. The conditioning period prior to test shall be 48 hours for specimens of 1/8 inch or less in thickness and 96 hours for thicker specimens.

4.5.1.2.2 Wet condition. Wet condition shall be two hours immersion in boiling, distilled water. The specimens shall be cooled to room temperature in the same water. The tests shall be made immediately after removal of the specimen from the water.

4.5.1.3 The test specimens shall be modified from Standard FTMS No. 406 to the following:

4.5.1.3.1 Tensile, circumferential. Specimens shall be in accordance with ASTM-D-2291-64T. Wall thickness shall be the nominal wall thickness of the tube; inside diameter shall conform to the part inside diameter and a width of two inches. Test specimens shall be tested in accordance with 4.5.4.

4.5.1.3.2 Tensile, longitudinal. Specimens shall be machined as shown in Figure 1.

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4.5.1.3.3 Flexural strength and modulus, circumferential and longitudinal. Specimens shall be one-half inch, nominal wall thickness, two inches wide, and ten inches long.

4.5.2 Determination of cure

4.5.2.1 Flexural strength method. Five post-cured specimens shall be prepared by heating them at 220°F for 24 hours and then cooling them in an air convection oven to room temperature. The flexural test method shall be that specified by Table I for flexural strength under the standard condition.

4.5.2.2 Barcol hardness method. Hardness of the laminate shall be determined by using a Barcol impressor instrument manufactured by Barber-Colman Company, or its equivalent. Measurements shall be made on surfaces which do not have gel coat or overlay by a method which conforms to the following: Hardness is measured by the depth of indentation made by a hardened steel truncated cone with an included angle of 26 degrees with a flat tip of 0.0062 inch in diameter. The penetrator fits into a hollow spindle and is held down by a spring loaded plunger. The depth of penetration under the spring-loaded pressure is transmitted to a dial indicator by a lever. The hardness scale on the dial has 100 divisions, each representing a depth of about 6.4 microns. In making the test, the leg plate in the rear of the instrument is set on the surface to be tested and a pressure of about 16 pounds applied against the point by pressing on the instrument housing; the hardness number is then read.

4.5.3 Percent voids. The percent voids shall be calculated from samples cut from at least five areas in the plastic unit as designated by the Government inspector (see Table I).

4.5.3.1 Method of calculation for clear unfilled laminate:

$$\text{Percent voids} = 100 - x$$

$$x = \frac{a(d)}{c} \text{ plus } \frac{a(e)}{b}$$

Where:

x = total calculated volume of laminate

a = specific gravity of laminate (Method 5011 of Standard FTMS No. 406) x 100 = weight of 100 cubic centimeters of laminate.

b = 2.57 (specific gravity of glass)

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c = specific gravity of cured resin (obtained from resin manufacturer or determined in accordance with Method 5011 of Standard FTMS No. 406).

d = resin content (expressed as a decimal) (determine in accordance with Method 7061 of Standard FTMS No. 406).

e = glass content (expressed as a decimal) = (1-d).

4.5.3.2 Method of calculation for filled laminate:

$$\text{Percent voids} = 100 - x$$

$$x = \frac{a(d)}{c} \text{ plus } \frac{a(e)}{b} \text{ plus } \frac{a(f)}{g}$$

Where:

x,a,b,c and d are the same as in 4.5.3.1

e = glass content (expressed as a decimal) = (1-d-f).

f = filler content (obtain from resin supplier).

g = specific gravity of filler (obtain from filler supplier).

4.5.4 Tensile, circumferential. The tensile strength shall be determined on specimens defined in 4.5.1.3.1. A ring jig to take this specimen, in accordance with ASTM-D-2290-64T, shall be employed to perform the test. Cross head travel shall be 0.20 to 0.25 inch per minute. Ultimate tensile strength shall be calculated as follows:

$$S_T = \frac{P}{2WT}$$

Where:

S_T = Ultimate tensile strength

P = Total load in pounds

W = Width in inches

T = Thickness in inches

4.5.5 Flexural strength, longitudinal and circumferential. Flexural strength (both directions) shall be determined using specimens described in 4.5.1.3.3 and the procedure defined in Standard FTMS No. 406, Method 1031.

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5. PREPARATION FOR DELIVERY

5.1 The packaging, packing, and marking requirements shall be as specified in the specification, drawings, or contracts for the parts fabricated from the plastic material.

6. NOTES

6.1 Intended use. The material described by this specification is intended for use in fabricating torpedo tubes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

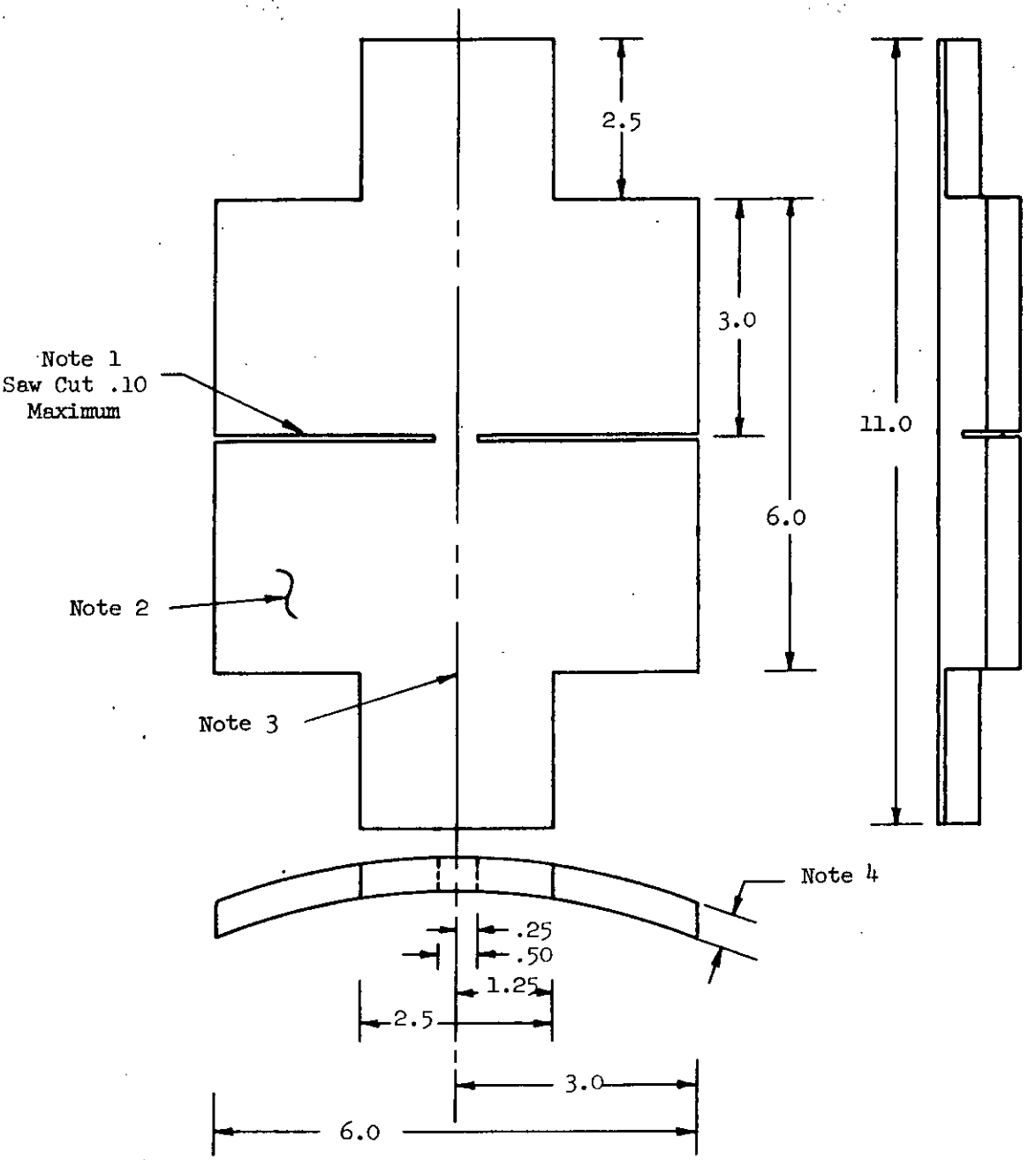
(b) Delivery schedule for preproduction (see 3.1) and production units.

(c) Packaging requirements.

(d) Other requirements concerning defects, desired surface quality, performance, etc., as necessary.

(e) Whether tooling is to be Government furnished, whether it shall become Government property, and whether it should be of a temporary nature or be capable of being shipped and reused following completion of the contract.

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Notes

1. Saw cut is perpendicular to centerline of sample.
2. Size of ears may be changed to fit tensile machine.
3. Centerline of sample will be parallel to longitudinal central axis of tube.
4. Sample thickness will be the wall thickness of tube.

Figure 1. Longitudinal Tensile Specimen.

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Naval Ordnance Systems Command
Washington, D.C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-P-82540(OS) PLASTIC MATERIALS, POLYESTER RESIN, Glass FIBER BASE, FILAMENT WOUND TUBE

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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